



Definition of Temporary Table in SQL

A **Temporary Table** in SQL is a special table that stores data temporarily during a database session. It is mainly used to hold intermediate results, simplify complex queries, and improve query organization during analysis or processing.

Temporary tables are automatically removed when:

- The session ends, or
- The database connection closes.

Key Characteristics of Temporary Tables

Feature	Description
Temporary Storage	Exists only for current session
Session-Based	Only visible to the user/session that created it
Faster Processing	Useful for intermediate calculations
Improves Readability	Breaks complex queries into smaller steps
Useful for Analytics	Common in ETL, dashboards, reporting

Short Notes and Explanation of Each Script

SECTION 1: INTRODUCTION TO TEMPORARY TABLES

```
USE techsphere_solutions;
```

What It Does

- Selects the techsphere_solutions database.
- Ensures all queries run inside the correct database.

Documentation

This is always the first step before querying tables.



SECTION 2: CREATING A TEMPORARY TABLE MANUALLY

```
CREATE TEMPORARY TABLE temp_project_assignments
(
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    project_name VARCHAR(100)
);
```

What It Does

Creates a temporary table named: temp_project_assignments.

Columns Created

- first_name
- last_name
- project_name

Documentation

This method manually defines the table structure before inserting data.

```
SELECT *
FROM temp_project_assignments;
```

What It Does

Displays all records from the temporary table.

Documentation

Initially, the table is empty because no data has been inserted yet.

SECTION 3: INSERTING DATA INTO TEMPORARY TABLE

```
INSERT INTO temp_project_assignments
VALUES('Benard', 'Omoga', 'AI Dashboard System');
```



What It Does

Inserts one record into the temporary table.

Documentation

Demonstrates how temporary tables can temporarily store custom or intermediate data.

```
SELECT *  
FROM temp_project_assignments;
```

What It Does

Retrieves the inserted temporary data.

Result

Shows:

- Employee name
- Assigned project

SECTION 4: CREATING TEMP TABLE FROM QUERY

```
CREATE TEMPORARY TABLE high_salary_roles  
SELECT  
    role_id,  
    role_title,  
    base_salary  
FROM job_roles  
WHERE base_salary > 100000;
```

What It Does

Creates a temporary table directly from a query result.

Result

Stores only:

- High-paying job roles
- Salaries above 100,000



Documentation

This is the fastest and most commonly used approach for creating temporary tables.

SECTION 5: VIEW TEMP TABLE CONTENT

```
SELECT *  
FROM high_salary_roles;
```

What It Does

Displays all records stored in high_salary_roles.

Documentation

Useful for quickly analyzing filtered datasets.

SECTION 6: TEMP TABLE WITH EMPLOYEE ANALYTICS

```
CREATE TEMPORARY TABLE employee_salary_analysis  
SELECT  
    e.employee_id,  
    CONCAT(e.first_name, ' ', e.last_name) AS employee_name,  
    d.department_name,  
    jr.role_title,  
    jr.base_salary
```

What It Does

Creates a temporary analytical dataset by joining multiple tables.

Combined Data Includes

- Employee names
- Departments
- Job roles
- Salaries

Documentation

This demonstrates how temp tables simplify complex reporting queries.



SECTION 7: QUERYING TEMP ANALYTICS TABLE

```
SELECT *  
FROM employee_salary_analysis  
ORDER BY base_salary DESC;
```

What It Does

Displays employees ordered from highest salary to lowest.

Documentation

Useful for payroll analysis and management reporting.

SECTION 8: BUSINESS ANALYTICS EXAMPLE

```
CREATE TEMPORARY TABLE department_salary_summary  
SELECT  
    d.department_name,  
    COUNT(e.employee_id) AS total_employees,  
    AVG(jr.base_salary) AS average_salary,  
    SUM(jr.base_salary) AS total_department_salary
```

What It Does

Creates a temporary salary summary report grouped by department.

Calculations Included

- Number of employees
- Average salary
- Total department salary

Documentation

Useful for:

- HR analytics
- Payroll planning
- Budget analysis



```
SELECT *  
FROM department_salary_summary;
```

What It Does

Displays department salary summary results.

Documentation

Provides management-level salary insights.

TEMP TABLE VS CTE

Temporary Table	CTE
Exists during session	Exists during query execution only
Can be queried multiple times	Used once in query
Stores physical temporary data	Virtual query result
Better for large workflows	Better for readability

Real-World Business Uses

Temporary tables are widely used for:

- Payroll processing
- Dashboard preparation
- Data transformation
- ETL pipelines
- Financial analysis
- KPI calculations
- HR reporting
- Large analytical workflows



Final Summary

This script demonstrates:

- Creating temporary tables manually
- Creating temp tables from queries
- Storing intermediate analytical data
- Querying temporary datasets
- Performing payroll and department analysis
- Using temp tables in business analytics workflows

It also shows how temporary tables help organize complex SQL operations into manageable and reusable steps.